

## STUDIES UPON THE COPPER FUNGICIDES

### I. THE INTERACTION OF COPPER SULPHATE WITH CALCIUM HYDROXIDE

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(With 4 Text-figures.)

NUMEROUS sprays in which the toxic action of copper derivatives upon fungi is utilised for the control of plant diseases caused by copper-sensitive fungi have been proposed but few have met with success. For most fungi against which these materials are employed, mycological data are available to place factors such as correct time and frequency of application upon a scientific basis. It is difficult therefore to attribute failure to incorrect use, and it is probable that success is due to chemical and physical factors, to examine which is the main purpose of the work to be described in this series. For knowledge of the chemical nature of the successful copper fungicides, of the changes the copper-containing deposit upon the plant undergoes after spraying and of the mechanism whereby fungus attack is prevented is scarce and to those questions which have been examined, various hypotheses supply conflicting and unsatisfactory answers.

Attention was given in the first place to Bordeaux mixture, for this, the oldest of the copper-containing sprays, has survived in spite of the inconveniences of its preparation. Bordeaux substitutes in paste or powder form have often been tried, but all appear to lack some of the chemical and physical properties which go to make the efficiency of Bordeaux mixture. To determine these properties it is necessary to ascertain the chemical character of the precipitate of Bordeaux mixture—the product of the interaction of copper sulphate and calcium hydroxide solutions.

#### THE PRECIPITATION OF CUPRIC SULPHATE SOLUTIONS BY CALCIUM HYDROXIDE.

The generally accepted views upon the composition of the precipitate of Bordeaux mixture are derived almost entirely from the classical research of Pickering (18). He showed that the composition of the precipitate was dependent upon the amount of calcium hydroxide added per

unit of copper sulphate, and concluded that a series of basic copper sulphates and basic copper-calcium sulphates were produced as the proportions of lime were increased. In the mixture prepared from equal weights of bluestone and quicklime, the precipitate had the composition  $10\text{CuO} \cdot \text{SO}_3 \cdot 4\text{CaO} \cdot \text{SO}_3$ . When smaller proportions of lime were used, he concluded that the precipitate consisted of less basic sulphates in which the ratios  $\text{CuO} : \text{SO}_3$  were 10 : 1, 5 : 1 and 4 : 1. Pickering's work has been repeated by two later investigators, Sicard<sup>(22)</sup> and Wöber<sup>(26)</sup>, both of whom have differed from him on important points. There is a general agreement, however, that the least basic sulphate contains the complex  $4\text{CuO} \cdot \text{SO}_3$  and is formed by the addition of calcium hydroxide in amounts up to 0.75 equivalent. The further addition of lime, according to Pickering, renders the supernatant solution alkaline to phenolphthalein, but until the molecular ratio of copper sulphate to calcium hydroxide reaches 10 : 9, this alkalinity disappears on standing. In his opinion this change in alkalinity was due to the interaction of the excess calcium hydroxide with the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate to form more basic compounds in which the ratio  $\text{CuO} \cdot \text{SO}_3$  was equal to 5 : 1 and 10 : 1. Sicard, who employed similar methods, found that the point of permanent alkalinity was not reached until an equimolecular amount of calcium hydroxide had been added, at which point he found that the composition of the precipitate agreed with the formula  $5\text{CuO} \cdot \text{SO}_3 \cdot \text{CaO}$ . Wöber, on the other hand, found that permanent alkalinity is produced if lime be added in excess of the ratio  $\text{CuO} : \text{CaO} = 10 : 8$ , the precipitate then corresponding to the formula  $\text{CuSO}_4 \cdot 4\text{Cu}(\text{OH})_2 \cdot \text{H}_2\text{O}$ . In support of this conclusion he showed that the formation of calcium sulphate ceased at this point.

These investigators employed stoichiometrical methods for the establishment of the identity of the compounds they considered to be formed. Since Pickering's time the conception of a chemical compound has been profoundly modified by the application of the phase rule. Further, the introduction of electrometric methods of determining hydrogen-ion concentration has not only provided a better method for the investigation of the reaction of the solution but has been used to throw light upon the phenomena of basic salt formation. For the particular problem of the chemistry of the Bordeaux precipitate belongs to the general problem of basic salt formation. It is now generally realised that the addition of alkali to solutions of metallic salts rarely results in the simple precipitation of the hydroxide. In most cases the precipitate contains a certain amount of the acidic radicle, carried down either by

adsorption or in the form of definite basic salts. The action of alkalis upon solutions of cupric salts has been employed frequently since Pickering's work for the study of this phenomenon, but, as this work is concerned mainly with the action of sodium hydroxide, it will not be discussed here except when, because of the analogy between this reaction and that of calcium hydroxide upon cupric salts, it throws light upon the chemistry of Bordeaux mixture.

For the study of the action of calcium hydroxide upon dilute copper sulphate solutions it was, however, found convenient to employ methods similar to those of Pickering, Sicard and Wöber, to interpret the results obtained after the manner of these workers and to criticise these interpretations from the view-point of modern physico-chemical theory. The general procedure has been to investigate the changes which occur when increasing amounts of lime water are added to solutions of copper sulphate, with special reference to the amount of sulphate radicle remaining in solution and to the hydrogen-ion concentration of the supernatant solution.

(1) *The amount of sulphate radicle present in solution.*

Details of the experimental work are given on p. 112 and from the data there recorded it is possible to draw a graph showing the percentage of sulphate radicle remaining in solution as the proportion of calcium oxide added is increased. Fig. 1 is derived from data obtained in Exp. A, Series 2. It will be seen that the first section of the curve shows a rapid decrease in the amount of sulphate radicle in solution to a point at which the amount of lime added corresponds to the ratio  $\text{CuO} : \text{CaO} = 1 : 0.75$ , at which point the whole of the copper initially present has been precipitated. If it be assumed that the precipitate consists of the  $4\text{CuO} \cdot \text{SO}_3$  basic salt the theoretical amount of sulphate radicle converted to calcium sulphate is indicated by the dotted line in the graph. That in many cases the individual determinations give points falling slightly below this line indicates a slight adsorption of the copper and calcium sulphates in solution by the precipitate. The analytical figures, given in Table II, for the amounts of copper and sulphate sulphur in the precipitate indicate that it is the  $4\text{CuO} \cdot \text{SO}_3$  basic salt.

The conclusion, that when an amount of calcium hydroxide less or equal to that which gives complete precipitation of the copper is added, the precipitate consists of the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate, is in agreement not only with Pickering, Sicard and Wöber, but also with the findings of those investigators who employed alkali hydroxide solutions instead of calcium hydroxide, *e.g.* Proust (21), Smith (23), Field (11) and Williamson (25).

More recently, however, Nelson(17) obtained indications that the initial compound precipitated by the addition, at ordinary temperatures, of sodium hydroxide to cupric sulphate solutions had a  $\text{CuO} : \text{SO}_3$  ratio of 3.5 : 1, a conclusion which might be explained if an adsorption of sulphate by the basic precipitate occurred. In view of the possibility, which Fowles(13) considered sufficient to invalidate the method, that alkali sulphate may be carried down by adsorption with the basic sulphate, electrometric methods have been applied to determine by indirect means

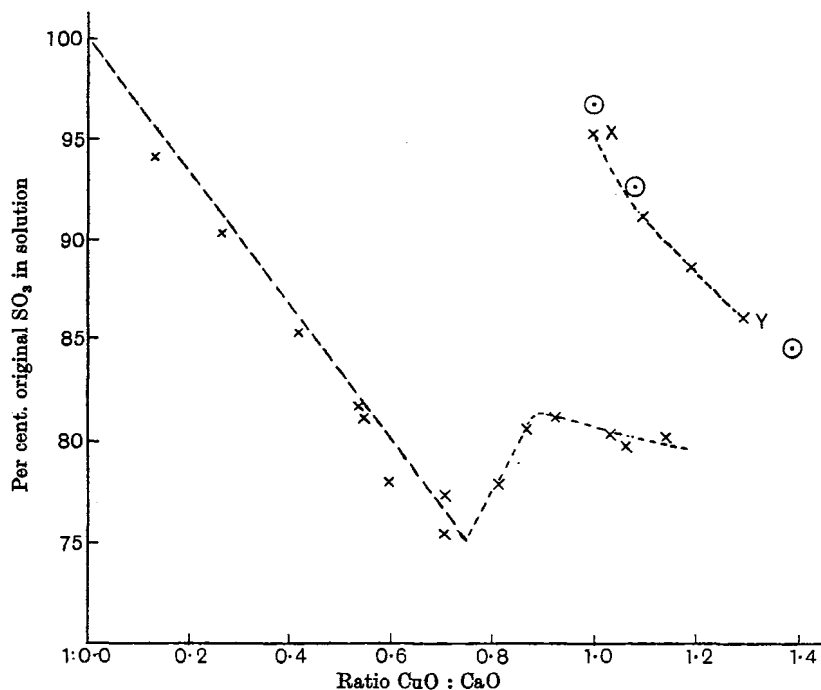


Fig. 1.

○ — ○ from data given by Bell and Cameron.  
 × — × data from Expt. A, Series 2.

the composition of the precipitate. If free copper sulphate be present the solution has an acid reaction due to hydrogen ions produced by the hydrolysis of the copper sulphate. At the point of complete precipitation of the copper there will therefore be a point of inflection upon the hydrogen-ion concentration curve, the position of which will not be affected by adsorption of alkali sulphate. The error introduced into the direct estimation of the  $\text{CuO} : \text{SO}_3$  ratio of the precipitate by the presence of alkali sulphate will thus be avoided. Employing an oxygen electrode,

Britton(4) found that if the sodium hydroxide be added slowly so that an amorphous precipitate is formed the hydrogen-ion concentration of the solution decreased rapidly after 1.47 equivalents of alkali had been added. He calculated that the precipitate contained amounts of copper and sulphate radicle in the molar ratio of 4 : 1. A similar conclusion is to be derived from both the quinhydrone and glass electrode studies quoted by Britton(7). Hopkins and Beebe(15) concluded that the single sharp inflection obtained with dilute solutions at room temperature indicated the formation of but one definite compound, the basic salt  $\text{CuSO}_4 \cdot 3\text{Cu}(\text{OH})_2 \cdot x\text{H}_2\text{O}$ .

The individuality of the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate has been indicated by phase rule studies of the system  $\text{CuO} - \text{SO}_3 - \text{H}_2\text{O}$ , but, owing to the time taken for equilibrium to be established, the results obtained by various workers are not in good agreement. Neither Bell and Taber(2) nor Young and Stearn(27) obtained indications of the formation of definite basic sulphates. Britton(5) concluded that, at  $25^\circ \text{C}$ ., there is but one basic sulphate which has the composition represented by the formula  $4\text{CuO} \cdot \text{SO}_3 \cdot 4\text{H}_2\text{O}$ . Bell and Murphy(1) also showed that, at  $100^\circ \text{C}$ ., the only basic salt formed has the composition in agreement with this formula. But in neither case does the plotting of the analytical data give a satisfactory point of intersection of the tie-lines and Bell and Murphy's conclusion is based upon the analysis of the washed precipitate. Posnjak and Tunell(20) studied the system at  $50^\circ$ ,  $100^\circ$  and  $200^\circ \text{C}$ . and found three basic sulphates:  $3\text{CuO} \cdot \text{SO}_3 \cdot 2\text{H}_2\text{O}$ ,  $4\text{CuO} \cdot \text{SO}_3 \cdot 3\text{H}_2\text{O}$  and  $3\text{CuO} \cdot 2\text{SO}_3 \cdot 5\text{H}_2\text{O}$ . The ranges of concentration over which the  $3\text{CuO} \cdot \text{SO}_3$  and  $3\text{CuO} \cdot 2\text{SO}_3$  basic salts appear increase with temperature. On the other hand, the  $4\text{CuO} \cdot \text{SO}_3$  salt, which does not appear in the  $200^\circ \text{C}$ . isotherm, is more prominent at the lowest temperature. Conclusions based upon the analogy between precipitation of copper sulphate solutions by sodium hydroxide and calcium hydroxide solutions involve the assumption that the function of the alkali is to remove the acid formed by the hydrolysis of the copper sulphate. Fowles(13) obtained evidence that the final product of hydrolysis of dilute cupric sulphate solutions is the 3 : 1 basic sulphate. This salt is presumably the  $3\text{CuO} \cdot \text{SO}_3 \cdot 2\text{H}_2\text{O}$  compound whose individuality was established by Posnjak and Tunell (*loc. cit.*). There is no evidence in the sulphate curve (Fig. 1) that compounds less basic than the  $4\text{CuO} \cdot \text{SO}_3$  salt are formed and a review of the literature shows that the majority of cases in which basic sulphates less basic than the 4 : 1 salt have been obtained were by precipitation from hot solution. This conclusion is

supported by the observation of Hopkins and Beebe<sup>(15)</sup> that, when the precipitate obtained by the addition of less than 1.5 equivalents of sodium hydroxide is boiled in the presence of copper sulphate, the 4 : 1 salt is slowly converted to a less basic sulphate which they suggested may be the 3 : 1 salt. Finally, the conclusion that the 3 : 1 salt is not formed by precipitation at room temperatures is supported by Fowles' <sup>(13)</sup> observation that the 3 : 1 salt is decomposed by cold water to form the 4 : 1 salt. Pickering's conclusion, that the 4 : 1 basic sulphate is precipitated by the addition of lime water in amounts less than 0.75 equivalent, is therefore confirmed by the present work and is substantiated, not only by recent work upon the analogous sodium hydroxide precipitation, but by phase rule studies of the system  $\text{CuO} - \text{SO}_3 - \text{H}_2\text{O}$ .

Continuing the study of the fate of the sulphate sulphur when amounts of calcium hydroxide in excess of 0.75 equivalents are added, it will be seen from Fig. 1 that part of the precipitated sulphate re-enters solution. This sulphate sulphur can only be formed by the decomposition of the 4 : 1 basic salt which, according to Pickering<sup>(18)</sup> and to Sicard<sup>(22)</sup>, is converted to more basic sulphates,  $5\text{CuO} \cdot \text{SO}_3$  and  $10\text{CuO} \cdot \text{SO}_3$ . Wöber<sup>(26)</sup> also considered that the appearance of further amounts of calcium sulphate followed conversion to the 5 : 1 salt, a conclusion supported by his observation that further amounts of calcium sulphate ceased to be formed when 0.8 equivalents of calcium hydroxide had been added. From the results shown in Fig. 1 it will be seen that there is a change of direction of the curve at the point when 0.9 equivalents of alkali had been added. The sulphate sulphur content of the precipitate at this point is approximately 20 per cent., and, applying Wöber's argument, it would be concluded that the 4 : 1 basic sulphate is converted to a compound of empirical composition  $5\text{CuO} \cdot \text{CaO} \cdot \text{SO}_3$ ; a basic copper-calcium sulphate for phase rule studies have shown no evidence of the existence of a  $5\text{CuO} \cdot \text{SO}_3$  basic sulphate in the system  $\text{CuO} - \text{SO}_3 - \text{H}_2\text{O}$ .

This conclusion involves the assumption that no adsorption of calcium sulphate by this compound has occurred. The failure of the sulphate radicle recovery to approach more closely to 100 per cent. might equally well be explained by adsorption and it is possible that the direction of the latter part of the curve is controlled mainly by the solubility of calcium sulphate in the supernatant solution. That the solubility of calcium sulphate may be the controlling factor is suggested by solubility data. The experimental determination of the solubility of calcium sulphate in calcium hydroxide solutions is made difficult by the tendency to form supersaturated solutions and it was deemed inadvisable to attempt the

estimation of solubility figures except by the direct method described on p. 113. If it be assumed that, for reasons which are discussed below, free calcium hydroxide will only appear permanently in solution after the ratio  $\text{CuO} : \text{CaO} = 1 : 1$  has been passed, the curve *XY* (Fig. 1) will represent the solubility of calcium sulphate in the calcium hydroxide solutions so produced.

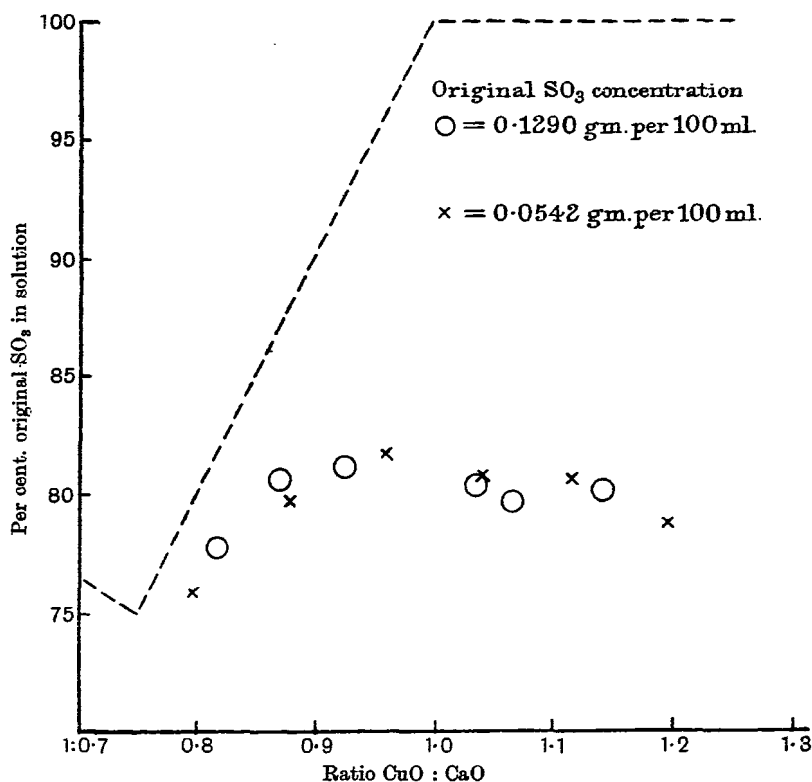


Fig. 2.

The disturbing factor introduced by the low solubility of calcium sulphate may be avoided by the use of more dilute solutions or by the examination of an analogous reaction in which a soluble calcium salt is formed. The results of a series of trials in which the initial concentration of the copper sulphate was 0.167 per cent. are illustrated in Fig. 2, the experimental data being recorded on p. 113. It will be seen that the amount of sulphate radicle in solution when amounts of lime water giving  $\text{CuO} : \text{CaO}$  ratio of between 1 : 0.75 and 1 : 1 are very similar to

those given in the first series of trials (Fig. 1), when the initial concentration of the copper sulphate was 0.25 per cent. This similarity would seem to give strong support to the view that the change of direction of the curve at the  $\text{CuO} : \text{CaO}$  ratio of 1 : 0.9 is due to the formation of a definite basic copper-calcium sulphate, but the extent of adsorption would have to be determined before any conclusion of its composition can be deduced from the analytical figures.

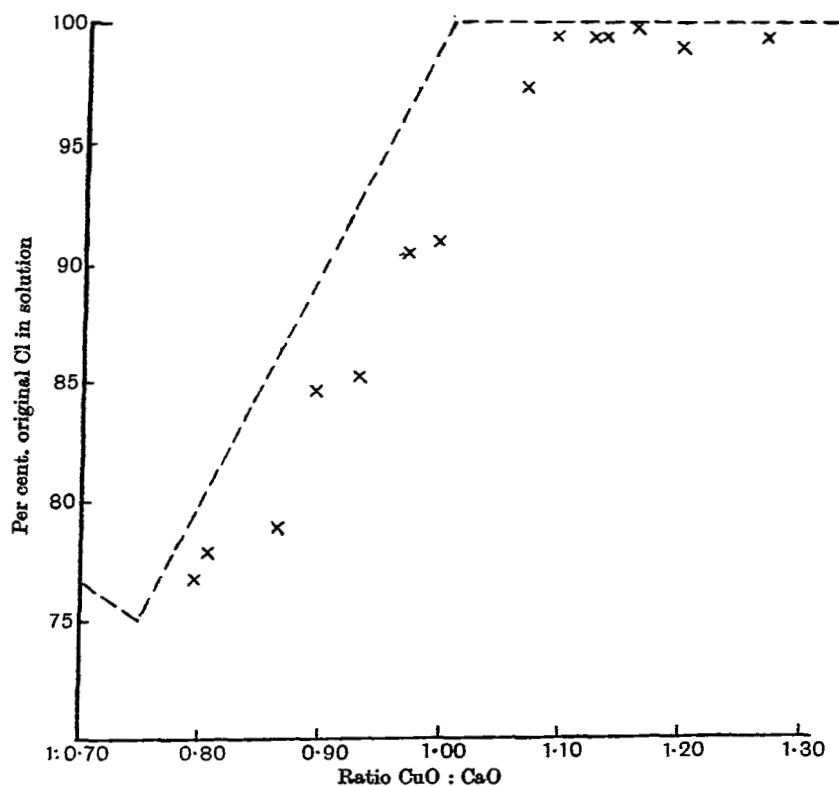


Fig. 3.

For the examination of an analogous reaction in which the calcium salt formed is of higher solubility than calcium sulphate, the interaction of copper chloride and calcium hydroxide was chosen. The results obtained are illustrated by Fig. 3, which incorporates the analytical figures given on p. 114. It is apparent that there is no break in the chloride content curve corresponding to that at 0.9 equivalents of calcium hydroxide in the sulphate curve. At 0.75 equivalents the basic chloride formed is the



4 : 1 salt and, if it be assumed that the reaction of this salt with additional calcium hydroxide involves the displacement of the chloride by the hydroxyl radicle, the theoretical amount of chloride in solution is given by the dotted line. That the actual value obtained falls below this line may be attributed to an adsorption by the precipitate of calcium chloride formed by this reaction.

This result suggests that the failure of the amount of sulphate radicle to reach nearer 100 per cent. recovery is due, not to the formation of an insoluble copper-calcium sulphate, but to adsorption of calcium sulphate by the compound produced by the interaction of lime water and the 4 : 1 basic sulphate. In dealing with the effect of adsorption of alkali sulphate upon the direct analytical estimation of the composition of the  $4\text{CuO} \cdot \text{SO}_3$  salt, it was pointed out that the indirect evidence afforded by the electrometric study of the reaction would avoid this error. In the same way an examination of the hydrogen-ion concentration of the supernatant solution as the ratio  $\text{CuO} : \text{CaO}$  approaches 1 : 1 may be used to give information of the composition of the precipitate and to avoid the error due to adsorption of alkali sulphate.

(2) *The hydrogen-ion concentration of the solution.*

Pickering (18) showed that the interaction of the  $4\text{CuO} \cdot \text{SO}_3$  salt with calcium hydroxide is indicated by changes in the hydrogen-ion concentration of the supernatant solution, as shown by the colour change of phenolphthalein. If this interaction be allowed time to attain equilibrium, the hydrogen-ion concentration curve should show inflections which serve to indicate the composition of the precipitate. If the change of direction of the sulphate sulphur curve at the ratio  $\text{CuO} : \text{CaO} = 1 : 0.9$  represents the formation of a basic copper-calcium sulphate, a point of inflection will occur on the  $p\text{H}$  curve at this point. If, as suggested by the study of the interaction of cupric chloride with lime water, the reaction between the 4 : 1 basic chloride and calcium hydroxide brings about a complete displacement of the chlorine from the basic chloride, the point of maximum change in  $p\text{H}$  will coincide with the ratio  $\text{CuO} : \text{CaO} = 1 : 1$ . Further, the formation of intermediate basic salts will be indicated by steps in the neutralisation curve which yield indirect information of the composition of such salts.

The electrometric studies of the interaction of alkalies and cupric salts by the investigators referred to in the previous section were confined to the action of sodium hydroxide. From the accounts of this work it would appear that definite alkalinity results after the addition of

1.50 equivalents of alkali. Britton (4), in his oxygen electrode titrations, showed that the  $pH$  of the solution rose rapidly to values approximating to 10 or 11 at this point, and he considered that Pickering's observation that permanent alkalinity was not reached until 1.8 equivalents had been added was due to the manner in which the alkali had been added. Hopkins and Beebe (15), who used the quinhydrone electrode, also obtained an abrupt increase in alkalinity after 1.5 equivalents had been added, the  $pH$  of the solution rising to above the point ( $pH$  8.5–9.0) at which errors are introduced into the quinhydrone determination through the neutralisation of the hydroquinone. Indicator methods, in the case of

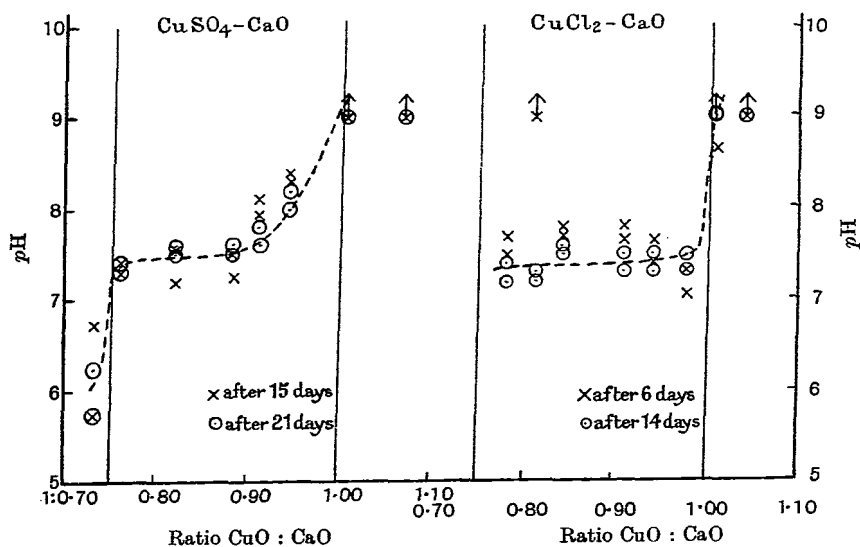


Fig. 4.

the calcium hydroxide reaction, led Pickering to consider the solution permanently alkaline at the ratio  $CuO : CaO = 10 : 9$ ; Sicard, 10 : 10; Wöber, 10 : 8.

Determinations of the hydrogen-ion concentration of the solution by means of the quinhydrone electrode by the method described on p. 115, are represented by the two graphs in Fig. 4. In the sulphate curve, the sudden increase in  $pH$  at the ratio  $CuO : CaO = 1 : 0.75$  indicates the formation of the  $4CuO \cdot SO_3$  basic sulphate and a less steep rise after 0.9 equivalents of lime water had been added does not give definite alkalinity until nearly 1 equivalent of calcium hydroxide is present. In the chloride curve the inflection at the ratio  $CuO : CaO = 1 : 1$  is

most marked and gives conclusive proof that the interaction of cupric chloride and calcium hydroxide involves equivalent amounts of the two reagents. In the same curve it will be noticed that, at the ratio  $\text{CuO} : \text{CaO} = 1 : 0.8$ , when only a slight excess of lime is present, its interaction with the basic chloride proceeds slowly and equilibrium was not reached by the sixth day. This phenomenon was observed also in Experiment E, Series 2, when the  $p\text{H}$  of the solutions shows a decrease with the amount of lime water added. Equilibrium is attained more rapidly when a larger excess of lime is present.

The conclusion that permanent alkalinity is produced only when amounts of lime in excess of the ratio  $\text{CuO} : \text{CaO} = 1 : 1$  are added is confirmed by the colour changes given with B.D.H. Universal indicator, recorded in Experiments A and C. In the cupric sulphate trials, two points of inflection on the neutralisation curve are clearly defined, the first, between the ratios  $\text{CuO} : \text{CaO} = 1 : 0.707$  and  $1 : 0.816$  ( $1 : 0.799$  in Experiment A, Series 3), corresponding to the complete conversion of the copper sulphate to the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate; the second, between  $\text{CuO} : \text{CaO}$  ratio of approximately  $1 : 1$ , indicating the complete decomposition of the  $4 : 1$  sulphate. Definite alkalinity, shown by the violet colour, only occurs with molar ratios  $\text{CuO} : \text{CaO}$  of over  $1 : 1$ . In the cupric chloride reaction there is a similar colour change extending over the ratios  $\text{CuO} : \text{CaO} = 1 : 0.996$  and  $1 : 1.087$ . The more alkaline colour at the ratio  $\text{CuO} : \text{CaO} = 1 : 0.808$  is further evidence of the slowness of equilibrium attainment when small amounts of lime water are added to the basic chloride suspension.

In the case of the cupric chloride reaction, the point of inflection on the neutralisation curve at the ratio  $\text{CuO} : \text{CaO} = 1 : 1$ , and the re-appearance of the whole of the precipitated chloride in solution, indicate that the reaction may be represented by the empirical equation:



The formation of the nigger-brown hydrated oxide is typical of the interaction of cupric salts with strong alkalies. A gelatinous blue precipitate is produced which, on standing, blackens more or less rapidly according to the experimental conditions, a change which is usually regarded as a process of dehydration. The blue precipitate which is initially produced in calcium hydroxide-cupric chloride reaction and which rapidly blackens, is of the same nature. Although it does not appear, from the literature, whether the blue precipitate is to be regarded as cupric hydroxide or as an hydrated oxide, it will be called for convenience

the hydroxide, an assumption which will not affect the subsequent argument.

The general similarity between the chloride and sulphate curves in Fig. 4 suggests that the action of calcium hydroxide upon the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate likewise results in the formation of cupric hydroxide. That the final insoluble product of each reaction is identical is indicated by the fact that the hydrogen-ion concentrations shown by the horizontal portions of the curves are similar; the  $[\text{OH}']$  components of the solubility products of the two precipitates are therefore equal.

The examination of the hydrogen-ion concentration curve indicates that the non-appearance of the total sulphate radicle in solution is due to adsorption. To postulate the existence of a basic copper-calcium sulphate to explain the change of direction of the sulphate in solution curve at the ratio  $\text{CuO} : \text{CaO} = 1 : 0.9$  becomes unnecessary and it follows that the precipitate, obtained when equimolecular amounts of copper sulphate and lime water are mixed, is cupric hydroxide upon which calcium sulphate is strongly adsorbed. The initial product of the reaction is the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate which, owing to its slight hydrolysis, slowly reacts with the excess calcium hydroxide to give cupric hydroxide. It thus becomes necessary to explain the fact that the cupric hydroxide so produced does not undergo the dehydration which occurs with the cupric chloride-lime water precipitate.

### (3) *The dehydration of the precipitate.*

Dehydration to form the brownish black cupric oxide occurred with the precipitate produced by the interaction of cupric chloride and amounts of lime water within the molar ratios  $\text{CuO} : \text{CaO} = 1 : 0.863$  and  $1 : 1.062$  in Experiment C,  $1 : 0.849$  and  $1 : 1.044$  in Experiment D, Series 2. The corresponding mixtures of copper sulphate and lime water showed no sign of darkening. Pickering<sup>(19)</sup>, however, observed that on boiling the precipitate produced from copper sulphate and lime water at ratios of  $\text{CuO} : \text{CaO}$  of between  $1 : 0.833$  and  $1 : 1.15$  decomposed, a fact which he considered evidence of the formation of an unstable  $5\text{CuO} \cdot \text{SO}_3$  basic sulphate. That dehydration does occur, though at  $100^\circ \text{C}.$ , in the sulphate reaction at the same  $\text{CuO} : \text{CaO}$  ratios as with the chloride reaction might also indicate the similarity of the two reactions. The chief difference in chemical character between the two precipitates is that the Bordeaux mixture precipitate contains much adsorbed calcium sulphate and it seems probable that this factor is responsible for the retention of the blue colour by the precipitate at ordinary temperatures. Retardation of the

dehydration of cupric hydroxide in the presence of sulphate ions was observed by Finch (12) and by Weiser (24). Confirmatory evidence of the stabilising action of adsorbed sulphate was obtained in Experiment E. The addition of calcium sulphate to the cupric chloride-calcium hydroxide mixture has prevented the blackening of the precipitate observed in the absence of the sulphate.

It follows that, if the copper sulphate-lime water precipitates obtained with molar  $\text{CuO} : \text{CaO}$  ratios greater than 1 : 0.75 can be freed from adsorbed sulphate and from excess of lime, they should undergo dehydration. Trials were therefore undertaken to discover the effect of prolonged washing with water upon the various basic precipitates.

The action of water upon the basic sulphate and upon the precipitate of Bordeaux mixture does not appear to have previously been studied, though it has often been assumed that, for the purposes of analysis, these materials can be freed from soluble impurities by washing with carbon dioxide-free water. In the case of the copper sulphate-calcium hydroxide precipitation both Sicard (22) and Wöber (26) washed the  $4\text{CuO} \cdot \text{SO}_3$  sulphate prior to analysis. Young and Stearn (27), Bell and Murphy (1) and Nelson (17) likewise washed the basic sulphate produced by the precipitation of copper sulphate with sodium hydroxide. Nelson added sodium hydroxide in excess of 1.5 equivalents yet washed the precipitate free from soluble sulphate upon the tacit assumption that the precipitate was not decomposed by water. Hooker (14), on the other hand, stated that the action of water upon the basic sulphate produced by adding sodium hydroxide insufficient in amount to cause complete precipitation of copper from copper sulphate resulted in the formation of a blue hydroxide. In Experiment F, Series 1, continued washing of the  $4\text{CuO} \cdot \text{SO}_3$  precipitate obtained by adding less than 0.75 equivalents of calcium hydroxide to copper sulphate, showed that this salt is stable, confirming Fowles' (13) statement.

With the more alkaline Bordeaux mixtures there are a few isolated references to the decomposition of the precipitate on washing. Wöber mentioned that prolonged washing with carbon dioxide-free water of the precipitate obtained with  $\text{CuO} : \text{CaO}$  ratios greater than 1 : 1, a complete removal of calcium compounds was not possible without decomposition of the precipitate. McCallan (16) also referred to the disorganisation of the precipitate of Bordeaux mixture when washed with large quantities of water. An examination of the effects of washing with carbon dioxide-free water, the precipitate obtained by the addition of more than 0.75 equivalents of calcium hydroxide to copper sulphate solutions is described in Experiment F, Series 2 (p. 116). In such cases complete or partial

dehydration of the precipitate occurred. Even with the "equal lime" Bordeaux mixture (Experiment F, Series 3) dehydration of the precipitate was produced by continued washing with water.

This result has an important practical bearing for, when sprayed upon foliage, the Bordeaux mixture precipitate will be subjected to continued washing by rain and dew. That, under these conditions, dehydration does not occur is shown by the retention of the blue colour of the spray deposit. The precipitate, in drying on the leaf, has therefore undergone some physical or chemical change whereby dehydration is prevented. That the physical change brought about by drying the precipitate is unable to prevent blackening is shown in Experiment G. Various Bordeaux mixtures were, in this experiment, allowed to stand in a Young's rod and disc fractional distillation tube arranged in a horizontal position. The supernatant liquid was drawn off and the precipitate dried by the continued passage of air through the tube. An adherent Bordeaux deposit was thus obtained which could be washed with water. In those trials in which atmospheric carbon dioxide was excluded blackening of the precipitate occurred. But when no precaution was taken to exclude carbon dioxide a Bordeaux deposit was obtained which was unchanged in colour by continued washing.

It must be concluded therefore that the retention of the blue colour by the Bordeaux precipitate upon foliage is due to chemical changes in which carbon dioxide is involved. The investigation of the action of carbon dioxide upon the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate and upon copper hydroxide which is formed from this compound by the action of lime will, it is hoped, throw light upon the nature of this reaction. A discussion of the corollaries of this conclusion must be held over until the nature of the deposit formed after spraying has been elucidated but there are two points which may be dealt with briefly at this stage.

That the Bordeaux deposit is formed *in situ* upon the leaf by the action, primarily, of carbon dioxide upon the copper sulphate-lime precipitate accounts for the excellent adhesiveness of the deposit. It has frequently been observed (see Boyd<sup>(3)</sup>) that a common cause of failure of substitutes for Bordeaux mixture is the rapid loss, from the foliage, of the protective copper-containing film. It is probable that in many cases this lack of adhesiveness is associated more with the chemical inertness of the substitute towards carbon dioxide, leaf excretions and other reactants which may come into play upon the foliage, than with the physical character of the substitute. In the case of Bordeaux mixture, many investigators have stressed the importance of the study of the physical nature of the precipitate. De Ong and Root<sup>(10)</sup> used the rate

of settling, whilst Butler (8) and others have employed the voluminousness of the precipitate as criteria for the adhesiveness of the precipitate. The value of such tests as indicators of the fungicidal efficiency of the spray may be profoundly affected by the susceptibility of the precipitate to these chemical changes which occur after spraying.

#### EXPERIMENTAL.

*Exp. A.* Definite volumes of lime water were added rapidly from an automatic pipette to flasks containing 25 ml. 4 per cent. A.R. copper sulphate ( $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ) and sufficient boiled-out distilled water to make the final volume up to 250 ml. The flasks, closed with a rubber stopper carrying a soda-lime tube to exclude atmospheric carbon dioxide, were placed in a thermostat at  $21.5^\circ \text{C}$ . In a number of the trials carbon dioxide was excluded by passing through the flasks carbon dioxide-free nitrogen, a pressure slightly greater than atmospheric being maintained within the flasks throughout the trial. The flasks were shaken from time to time and after the lapse of a definite period—seven or twenty-eight days—were removed from the thermostat, allowed to stand overnight and the contents sampled.

For analysis 100 ml. of the supernatant liquid, withdrawn by means of a pipette, were used for the estimation of calcium in solution. The contents of the flask were then poured on a rapid filter and, without delay, 100 ml. of the filtrate were taken for the estimation of sulphate and copper content. 20 ml. of the filtrate were also used for an estimate of hydrogen-ion concentration with the B.D.H. Universal indicator. The precipitate, with the remaining 30 ml. filtrate, was dissolved in the minimum of hydrochloric acid and the content of calcium, copper and sulphate determined.

Table I.

*Series 1. Mixtures contained 0.3225 gm.  $\text{SO}_3$  per 250 ml., and were left in thermostat for 7 days.*

| Lime water<br>added gave<br>molar ratio<br>$\text{CuO} : \text{CaO}$ | $\text{SO}_3$ in solution |               | Colour with B.D.H.<br>Universal indicator | In precipitate    |                  |                  |
|----------------------------------------------------------------------|---------------------------|---------------|-------------------------------------------|-------------------|------------------|------------------|
|                                                                      | gm.                       | % of<br>total |                                           | $\text{SO}_3$ gm. | $\text{CuO}$ gm. | $\text{CaO}$ gm. |
| 1 : 0.133                                                            | 0.3077                    | 95.4          | Orange                                    | 0.0130            | 0.0577           | Nil              |
| 1 : 0.266                                                            | 0.2938                    | 91.1          | Orange                                    | 0.0303            | —                | —                |
| 1 : 0.417                                                            | 0.2780                    | 86.2          | Orange                                    | 0.0427            | 0.1827           | —                |
| 1 : 0.532                                                            | 0.2636                    | 81.7          | Orange                                    | 0.0602            | 0.2295           | Nil              |
| 1 : 0.696                                                            | 0.2502                    | 77.6          | Orange                                    | 0.0746            | —                | —                |
| 1 : 0.706                                                            | 0.2433                    | 75.4          | Orange                                    | 0.0773            | 0.2997           | 0.0011           |
| 1 : 0.899                                                            | 0.2557                    | 79.3          | Greenish yellow                           | 0.0693            | 0.3205           | 0.0200           |
| 1 : 0.963                                                            | 0.2574                    | 79.8          | Greenish yellow                           | 0.0643            | 0.3205           | 0.0309           |
| 1 : 1.064                                                            | 0.2557                    | 79.3          | Violet                                    | 0.0674            | 0.3205           | 0.0530           |
| 1 : 1.091                                                            | 0.2575                    | 79.8          | Violet                                    | 0.0648            | 0.3205           | 0.0527           |
| 1 : 1.155                                                            | 0.2565                    | 79.5          | Violet                                    | 0.0637            | 0.3205           | 0.0608           |

Table II.

*Series 2. Mixtures contained 0.3225 gm. SO<sub>3</sub> per 250 ml., and were left in thermostat for 28 days.*

| Lime water added gave molar ratio<br>CuO : CaO | SO <sub>3</sub> in solution |            | Colour with B.D.H. Universal indicator | In precipitate      |         |         | Molar ratio<br>CuO : SO <sub>3</sub> |
|------------------------------------------------|-----------------------------|------------|----------------------------------------|---------------------|---------|---------|--------------------------------------|
|                                                | gm.                         | % of total |                                        | SO <sub>3</sub> gm. | CuO gm. | CaO gm. |                                      |
| 1 : 0.133                                      | 0.3040                      | 94.0       | Orange                                 | 0.0158              | 0.0582  | Nil     | 3.7 : 1                              |
| 1 : 0.266                                      | 0.2912                      | 90.3       | Orange                                 | 0.0304              | —       | —       | —                                    |
| 1 : 0.417                                      | 0.2752                      | 85.3       | Orange                                 | 0.0457              | 0.1814  | —       | 4.0 : 1                              |
| 1 : 0.532                                      | 0.2636                      | 81.7       | Orange                                 | 0.0603              | 0.2315  | Nil     | 3.9 : 1                              |
| 1 : 0.545                                      | 0.2617                      | 81.1       | Orange                                 | 0.0591              | 0.2362  | Nil     | 4.0 : 1                              |
| 1 : 0.598                                      | 0.2520                      | 78.0       | —                                      | 0.0681              | 0.2657  | Nil     | 3.9 : 1                              |
| 1 : 0.696                                      | 0.2513                      | 77.9       | Orange                                 | 0.0708              | —       | —       | —                                    |
| 1 : 0.707                                      | 0.2429                      | 75.3       | Orange                                 | 0.0783              | 0.3095  | 0.0001  | 4.0 : 1                              |
| 1 : 0.816                                      | 0.2511                      | 77.8       | Yellowish green                        | 0.0734              | 0.3221  | 0.0082  | —                                    |
| 1 : 0.870                                      | 0.2599                      | 80.6       | Greenish yellow                        | 0.0623              | 0.3212  | 0.0184  | —                                    |
| 1 : 0.925                                      | 0.2615                      | 81.1       | —                                      | 0.0614              | 0.3205  | 0.0246  | —                                    |
| 1 : 1.034                                      | 0.2589                      | 80.3       | Bluish green                           | 0.0608              | 0.3198  | 0.0499  | —                                    |
| 1 : 1.064                                      | 0.2571                      | 79.7       | Violet                                 | 0.0648              | 0.3175  | —       | —                                    |
| 1 : 1.142                                      | 0.2586                      | 80.2       | Violet                                 | 0.0647              | 0.3205  | 0.0550  | —                                    |

Table III.

*Series 3. The final volume of the copper sulphate-lime water mixture was adjusted to 450 ml., 25 ml. 3 per cent. copper sulphate being used. The flasks were left for 28 days before sampling for analysis.*

| Lime water added gave molar ratio<br>CuO : CaO | In solution         |            | Colour with B.D.H. Universal indicator |
|------------------------------------------------|---------------------|------------|----------------------------------------|
|                                                | gm. SO <sub>3</sub> | % of total |                                        |
| 1 : 0.799                                      | 0.1875              | 76.9       | Greenish yellow                        |
| 1 : 0.879                                      | 0.1945              | 79.7       | Greenish yellow                        |
| 1 : 0.959                                      | 0.1993              | 81.6       | Greenish yellow                        |
| 1 : 1.039                                      | 0.1971              | 80.7       | Greenish yellow                        |
| 1 : 1.118                                      | 0.1966              | 80.6       | Blue                                   |
| 1 : 1.198                                      | 0.1922              | 78.7       | Violet                                 |

A final check upon the accuracy of the determinations was obtained by calculating from the analytical figures the total amounts of copper, calcium and sulphate radicle present. Unless this total agreed within 1 per cent. to the amounts of copper sulphate solution and lime water initially used, the figures were discarded.

*Exp. B.* 1 gm. amounts of calcium sulphate (B.D.H. pure precipitated) were weighed into 300 ml. flasks, various amounts of saturated lime water were added and the volume adjusted to 250 ml. with boiled-out distilled water. The flasks were closed with a rubber bung carrying a soda-lime tube to exclude carbon dioxide and were left, with intermittent



shaking for 5 weeks in a thermostat at 21.5° C. Aliquots of the clear supernatant liquid were then withdrawn for analysis. Calculation of the quantities of sulphate and of calcium per 100 ml. solution gave the following figures:

|                   |        |        |        |            |
|-------------------|--------|--------|--------|------------|
| SO <sub>3</sub> : | 0.1227 | 0.1175 | 0.1142 | 0.1110 gm. |
| CaO:              | 0.0845 | 0.0911 | 0.0978 | 0.1040 gm. |

The figures determined by Cameron and Bell<sup>(9)</sup> who studied the system CaSO<sub>4</sub> — CaO — H<sub>2</sub>O at 25° C. give the following figures:

|                   |        |        |                        |
|-------------------|--------|--------|------------------------|
| SO <sub>3</sub> : | 0.1250 | 0.1194 | 0.1090 gm. per 100 ml. |
| CaO:              | 0.0876 | 0.0898 | 0.1112 gm. per 100 ml. |

*Exp. C.* For the investigation of the interaction of cupric chloride and lime water, an experimental procedure similar to that employed in *Exp. A* was used, an equivalent solution of cupric chloride (27.5 gm. A.R. CuCl<sub>2</sub> · 2H<sub>2</sub>O per litre) replacing the copper sulphate solution. The flasks were placed in a thermostat at 21.5° C. for one week before sampling.

Table IV.

*Mixtures contained 0.2883 gm. CuO per 250 ml.*

| Lime water<br>added gave<br>molar ratio<br>CuO : CaO | Cl in solution |               | Colour with B.D.H.<br>Universal indicator | Colour of<br>precipitate |
|------------------------------------------------------|----------------|---------------|-------------------------------------------|--------------------------|
|                                                      | gm.            | % of<br>total |                                           |                          |
| 1 : 0.797                                            | 0.2214         | 76.8          | Orange                                    | Greenish blue            |
| 1 : 0.808                                            | 0.2246         | 77.9          | Green                                     | Greenish blue            |
| 1 : 0.863                                            | 0.2278         | 79.0          | Greenish yellow                           | Nigger brown             |
| 1 : 0.896                                            | 0.2439         | 84.6          | —                                         | Nigger brown             |
| 1 : 0.930                                            | 0.2461         | 85.3          | Greenish yellow                           | Nigger brown             |
| 1 : 0.959                                            | 0.2608         | 90.4          | —                                         | Trace of black           |
| 1 : 0.996                                            | 0.2621         | 90.9          | Bluish green                              | Bluish grey              |
| 1 : 1.062                                            | 0.2802         | 97.2          | Bluish green                              | Bluish grey              |
| 1 : 1.087                                            | 0.2866         | 99.4          | Violet                                    | Blue                     |
| 1 : 1.119                                            | 0.2866         | 99.4          | Violet                                    | Blue                     |
| 1 : 1.129                                            | 0.2866         | 99.4          | —                                         | —                        |
| 1 : 1.152                                            | 0.2877         | 99.8          | —                                         | —                        |
| 1 : 1.196                                            | 0.2856         | 99.0          | Violet                                    | Blue                     |
| 1 : 1.262                                            | 0.2866         | 99.4          | —                                         | —                        |

The estimation of chloride in solution when amounts of lime water less than the molar ratio CuO : CaO = 1 : 0.75 were added was frustrated by the nature of the precipitate produced. An opalescent greenish blue colloidal solution was formed which made it impossible to remove the basic chloride by filtration or by sedimentation. It has been observed by Britton<sup>(6)</sup> that the production of colloidal solutions of basic salts is a characteristic tendency of chlorides.

*Exp. D, Series 1.* Mixtures containing known amounts of copper sulphate and calcium hydroxide were prepared, as in *Exp. A*, by the

addition of definite volumes of lime water to 50 ml. 4 per cent. copper sulphate diluted to give a final volume of 500 ml. The solutions were protected from atmospheric carbon dioxide by covering with a layer of medicinal paraffin.

To determine the hydrogen-ion concentration of the supernatant liquid, about 5 ml. were withdrawn with a pipette and added to a small cup-shaped separating funnel containing solid quinhydrone and sufficient medicinal paraffin to give a protective layer over the solution. The presence of medicinal paraffin did not interfere with the determination except by reducing the sensitiveness of the platinum electrode—a result probably of the continued ignition of the trace of paraffin left on the electrode after washing. As the null point reading was not always sharply defined, a wide interpretation was given to the millivolt reading obtained. The ability of the paraffin layer to exclude carbon dioxide was shown by a preliminary trial in which a dilute solution of calcium sulphate, made slightly alkaline with 4 drops of lime water, was allowed to stand under paraffin. The initial *pH* of the solution was 8.75–8.85 and after standing 27 days the figure was reduced only to *pH* 8.55–8.60.

Table V.

*Series 1. Copper sulphate-lime water.*

| Lime water<br>added gave<br>molar ratio<br>CuO : CaO | <i>pH</i> of supernatant solution after standing |         |           |           |
|------------------------------------------------------|--------------------------------------------------|---------|-----------|-----------|
|                                                      | 5 days                                           | 7 days  | 15 days   | 21 days   |
| 1 : 0.735                                            | 5.6–6.0                                          | 5.6–6.0 | 5.75–6.75 | 5.75–6.25 |
| 1 : 0.765                                            | 6.5–6.8                                          | 6.9–7.2 | 7.3–7.4   | 7.3–7.4   |
| 1 : 0.826                                            | 7.0–7.2                                          | 7.5–7.6 | 7.2–7.55  | 7.5–7.6   |
| 1 : 0.887                                            | 7.0–7.2                                          | 7.6–7.7 | 7.25–7.5  | 7.5–7.6   |
| 1 : 0.918                                            | 7.8–7.9                                          | 8.0–8.1 | 7.95–8.15 | 7.6–7.8   |
| 1 : 0.949                                            | 8.7                                              | 8.4     | 8.3–8.4   | 8.0–8.2   |
| 1 : 1.010                                            | >9.0                                             | >9.0    | >9.0      | >9.0      |
| 1 : 1.071                                            | >9.0                                             | >9.0    | >9.0      | >9.0      |

*Series 2. Copper chloride-lime water.*

| Lime water<br>added gave<br>molar ratio<br>CuO : CaO | Colour of precipitate<br>after 2 days | <i>pH</i> of supernatant solution<br>after standing |         |
|------------------------------------------------------|---------------------------------------|-----------------------------------------------------|---------|
|                                                      |                                       | 6 days                                              | 14 days |
| 1 : 0.783                                            | Greenish blue                         | 7.5–7.7                                             | 7.2–7.4 |
| 1 : 0.816                                            | Greenish blue                         | >9.0                                                | 7.2–7.3 |
| 1 : 0.849                                            | Nigger brown                          | 7.6–7.8                                             | 7.5–7.7 |
| 1 : 0.914                                            | Nigger brown                          | 7.65–7.8                                            | 7.3–7.5 |
| 1 : 0.947                                            | Nigger brown                          | 7.4–7.65                                            | 7.3–7.5 |
| 1 : 0.980                                            | Nigger brown                          | 7.05–7.3                                            | 7.3–7.5 |
| 1 : 1.012                                            | Nigger brown                          | 8.65                                                | >9.0    |
| 1 : 1.044                                            | Blackish purple                       | >9.0                                                | >9.0    |
| 1 : 1.110                                            | Blue                                  | >9.0                                                | >9.0    |

*Exp. E.* A series of six cupric chloride-lime water mixtures was made up, in three of the six flasks saturated calcium sulphate solution replaced the water required to complete the volume to 250 ml. Since the retention of the blue colour at ratios CuO : CaO of over 1 : 1 is to be attributed to the alkalinity of the mixture, the hydrogen-ion concentration of the supernatant solution was determined.

Table VI.

| Lime<br>water<br>added<br>gave<br>molar<br>ratio<br>CuO : CaO | gm.<br>SO <sub>3</sub><br>pre-<br>sent | Hydrogen-ion<br>concentration<br>after |         | Colour of precipitate after          |                                     |                             |
|---------------------------------------------------------------|----------------------------------------|----------------------------------------|---------|--------------------------------------|-------------------------------------|-----------------------------|
|                                                               |                                        | 6 days                                 | 18 days | 1 day                                | 6 days                              | 18 days                     |
| 1 : 0.849                                                     | Nil                                    | 8.1-8.2                                | 7.5-7.7 | Greenish blue                        | Blackish blue                       | Black, some<br>bluish slate |
| 1 : 0.849                                                     | 0.110                                  | 7.2-7.5                                | 7.3-7.5 | Blue, with tinge<br>of greenish blue | Blue                                | Blue                        |
| 1 : 0.914                                                     | Nil                                    | 7.9-8.1                                | 7.3-7.5 | Greyish black                        | Nigger brown, some<br>blackish blue | Nigger brown                |
| 1 : 0.914                                                     | 0.098                                  | 7.2-7.5                                | 7.4-7.6 | Blue                                 | Blue                                | Blue                        |
| 1 : 0.979                                                     | Nil                                    | 7.0-7.4                                | 7.3-7.5 | Almost black                         | Nigger brown                        | Nigger brown                |
| 1 : 0.979                                                     | 0.087                                  | 7.2-7.4                                | 7.3-7.5 | Blue                                 | Blue                                | Blue                        |

*Exp. F, Series 1.* Two litres of lime water were added to a solution containing 10.50 gm. CuSO<sub>4</sub>.5H<sub>2</sub>O, precautions being taken to exclude carbon dioxide. The bottle was closed with a stopper carrying two tubes, one reaching almost to the bottom of the bottle to enable the withdrawal of the supernatant liquid by means of a filter pump, the second permitting air free from carbon dioxide to enter. By connecting the second tube to the filter pump, a quantity of freshly-boiled distilled water equal in volume to the liquid removed was added through the first tube. Washings were repeated when the precipitate had settled. After five washings the amount of calcium sulphate in the wash water was too small for analysis. The amounts of calcium oxide and of copper sulphate used were known and by subtraction of the total amounts of calcium oxide, copper and sulphate radicle removed in the wash water, the molar ratio CuO : SO<sub>3</sub> in the precipitate was calculated to be 4.05 : 1.

*Series 2.* Lime water was added to copper sulphate solutions prepared by diluting 50 ml. 4 per cent. copper sulphate with carbon dioxide-free water to amounts necessary to give a final volume of 450 ml. The mixtures were protected from atmospheric carbon dioxide by covering with a layer of medicinal paraffin. After standing 3 weeks the super-

natant solutions were withdrawn by syphoning and the volume restored by adding boiled distilled water. Every 12 hours the process was repeated, and, after the sixth washing, the mixtures were left for 48 hours. The colour changes of the precipitate are shown in Table VII.

Table VII.

| Lime water<br>added gave<br>molar ratio<br>CuO : CaO | Colour of the precipitate after |                                    |                                    |
|------------------------------------------------------|---------------------------------|------------------------------------|------------------------------------|
|                                                      | 4 washings                      | 5 washings                         | 6 washings                         |
| 1 : 0.735                                            | Greenish blue                   | Greenish blue,<br>partly colloidal | Greenish blue,<br>partly colloidal |
| 1 : 0.765                                            | Blue, with<br>greenish tint     | Blue, with<br>greenish tint        | Blue                               |
| 1 : 0.826                                            | Blue                            | Blue                               | Bluish slate                       |
| 1 : 0.887                                            | Blue                            | Signs of blackening                | Blackish blue                      |
| 1 : 0.918                                            | Signs of blackening             | Blackish grey                      | Nigger brown                       |
| 1 : 0.949                                            | Definite blackening             | Nigger brown                       | Nigger brown                       |
| 1 : 1.010                                            | Blue                            | Definite blackening                | Blackish grey                      |
| 1 : 1.071                                            | Blue                            | Definite blackening                | Blackish grey                      |

*Series 3.* Bordeaux mixture (8 : 8 : 100) was prepared by the addition of 80 ml. 10 per cent. copper sulphate solution to 920 ml. water containing 12 gm. hydrated lime. The mixture was allowed to settle and the supernatant liquid withdrawn by suction. An amount of carbon dioxide-free water equal in volume to the solution withdrawn was added and the mixture shaken. After settling, the process was repeated. After seven such treatments the bluish mauve precipitate began to darken in colour and, after ten washings, it was nigger brown in colour. After further washings the precipitate became partly colloidal and washing by decantation was impossible.

*Exp. G.* To obtain a deposit of Bordeaux mixture dried in the absence of carbon dioxide, the precipitate was transferred by suction to a Young's rod and disc fractional distillation tube, containing 15 discs, of which the thermometer opening was closed by a waxed rubber bung. The precipitate was allowed to settle and the supernatant liquid withdrawn by gentle suction, the delivery tube being connected to a train of three U-tubes filled with calcium chloride and three of soda lime. Air thus dried and freed from carbon dioxide was drawn over the deposit until all moisture had been removed. In a second distillation tube, placed alongside the first, no precautions were taken to exclude carbon dioxide. The rate of flow of air through the two tubes was similar.

An "equal lime" Bordeaux mixture was prepared by the addition of 40 ml. 3 per cent. crystalline copper sulphate to 1 litre of lime water.

The precipitate was allowed to settle and transferred to the Young's tubes. After drying, the colour of the deposit in the carbon dioxide-free tube was a purer and deeper blue than that dried without removal of carbon dioxide. The calcium chloride tubes were then disconnected and the Young's tubes connected with flasks containing boiled-out distilled water, a slow stream of water being drawn over the Bordeaux deposit by means of a filter pump. After the passage of 4 litres of water the deposit in the tube protected from carbon dioxide began to blacken just below the entrance tube and after 12 litres had passed (24 hours) the whole of the deposit in the tube was definitely blackened. The deposit in the tube, unprotected from carbon dioxide during drying, was still blue in colour and was apparently unaffected.

Similar results were obtained with Bordeaux mixtures prepared by the addition of copper sulphate solution to lime water in amounts giving CuO : CaO ratios of 1 : 1 and 1 : 2.

#### SUMMARY.

1. The initial product of the interaction of cupric sulphate and calcium hydroxide solutions, at ordinary temperatures, is the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate.

2. With amounts of calcium hydroxide in excess of 0.75 equivalent, the  $4\text{CuO} \cdot \text{SO}_3$  salt is slowly decomposed with elimination of the sulphate radicle and the formation of a blue hydrated cupric oxide or cupric hydroxide, a product called, for convenience, the hydroxide.

3. The cupric hydroxide formed retains, by adsorption, part of the sulphate radicle. Direct evidence by analysis of the precipitate, which might be considered to indicate the formation of a basic copper-calcium complex, is therefore unreliable.

4. In the analogous interaction of calcium hydroxide with cupric chloride, the formation of cupric hydroxide in the presence of amounts of calcium hydroxide in excess of 0.75 equivalent, is indicated by the dehydration of the precipitate to form nigger brown hydrated cupric oxide, and by the definite point of inflection on the hydrogen-ion concentration curve which corresponds to equimolecular amounts of calcium hydroxide and cupric chloride.

5. The hydrogen-ion concentration curve for the cupric sulphate-calcium hydroxide reaction shows that permanent alkalinity is not reached until equimolecular amounts of calcium hydroxide have been added.

6. The cupric hydroxide, formed from the  $4\text{CuO} \cdot \text{SO}_3$  basic sulphate by the action of calcium hydroxide, is stabilised by adsorbed sulphate ions. If the adsorbed sulphate and excess calcium hydroxide be removed by continued washing with carbon dioxide-free water, the Bordeaux mixture precipitate undergoes dehydration to form the nigger brown hydrated cupric oxide.

7. Dehydration of the Bordeaux mixture precipitate when sprayed upon foliage, is therefore prevented by some chemical or physical change of the precipitate which occurs on drying.

8. Dehydration is not prevented by physical changes which occur on drying.

9. Carbon dioxide is a factor in the process whereby dehydration is prevented.

10. It is suggested that the formation *in situ* of the protective copper-containing deposit by the action, primarily, of carbon dioxide, accounts for the excellent adhesive properties of the Bordeaux mixture precipitate; that inertness towards the chemical changes which occur after spraying may be the cause of failure of some Bordeaux substitutes; that these changes may profoundly affect the value of sedimentation tests as criteria for the fungicidal efficiency of various Bordeaux mixtures.

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